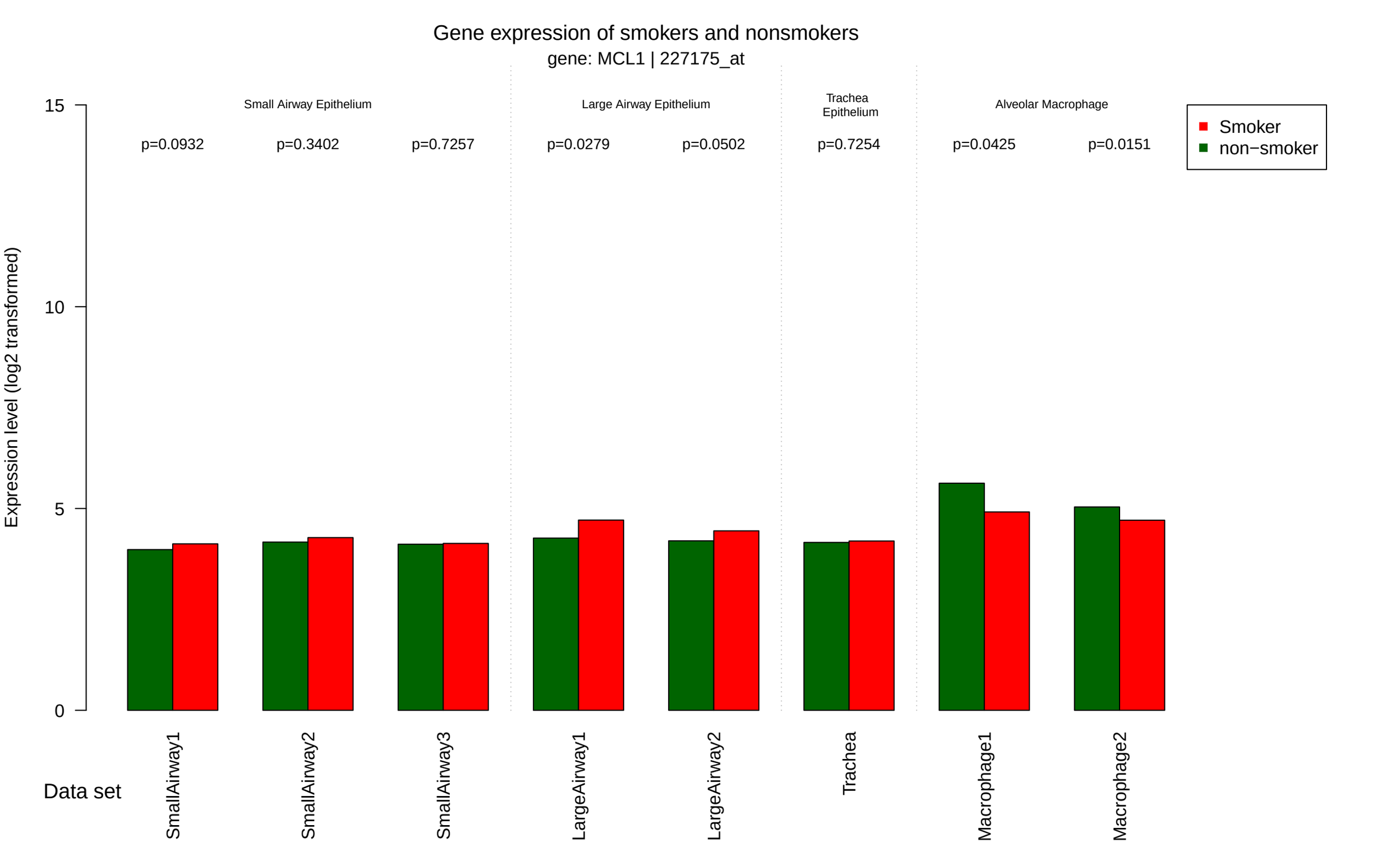
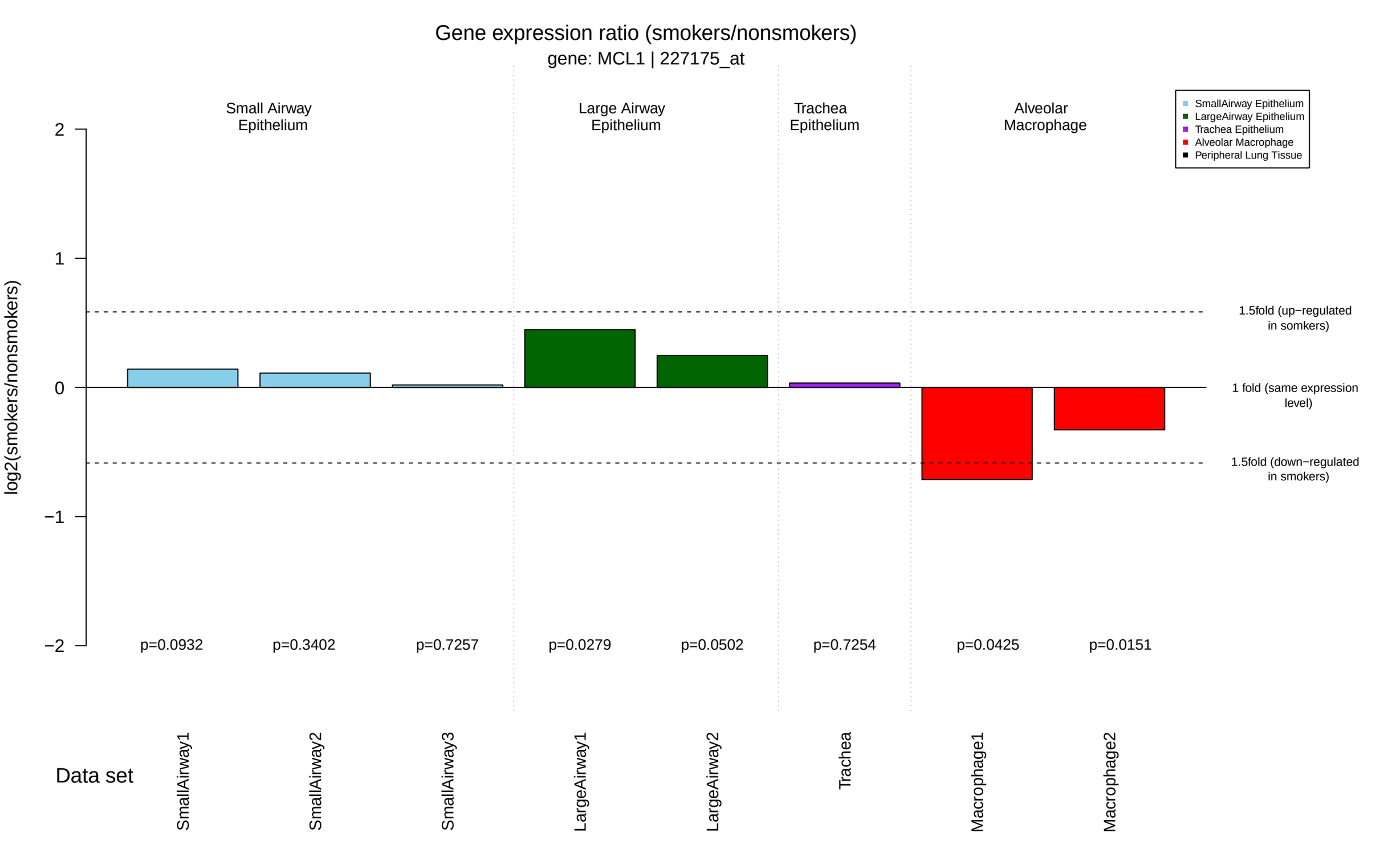
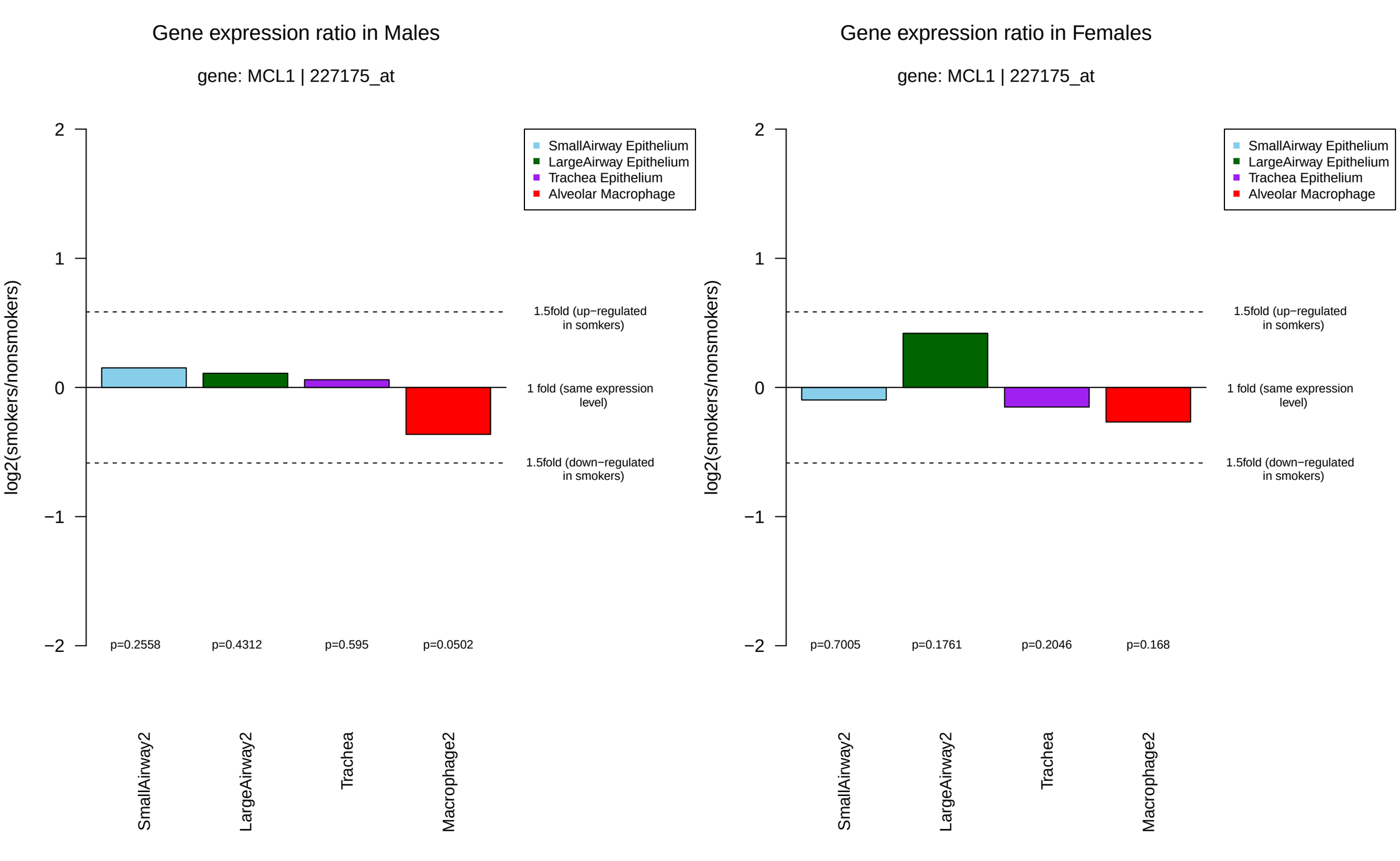
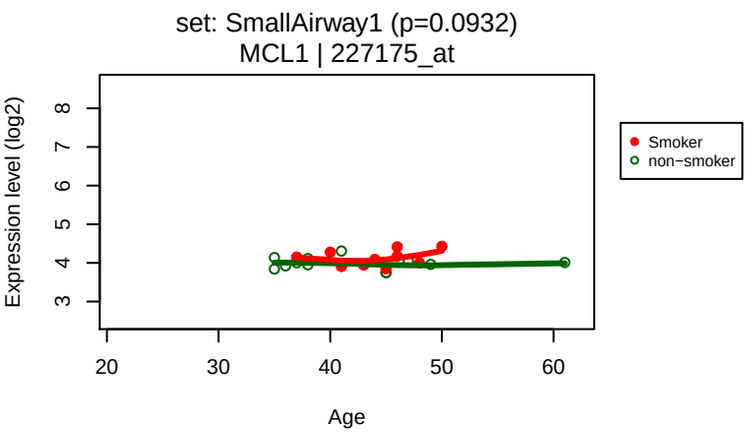
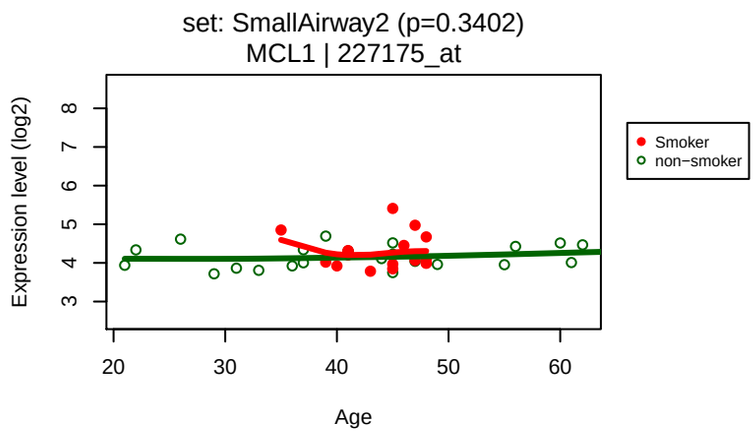
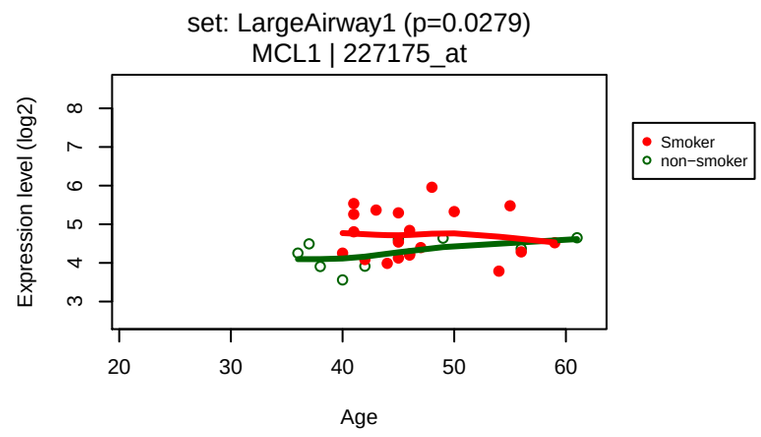

Gene expression of smokers and nonsmokers
gene: MCL1 | 227175_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0932
p=0.3402
p=0.7257
p=0.0279
p=0.0502
p=0.7254
p=0.0425
p=0.0151
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

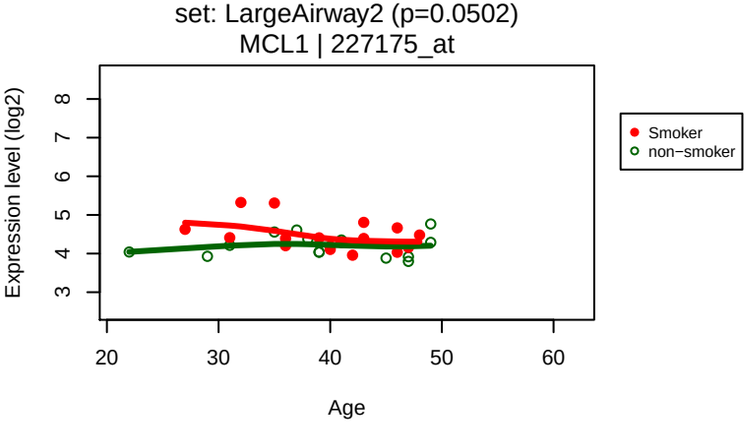

Gene expression ratio (smokers/nonsmokers)
gene: MCL1 | 227175_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0932
p=0.3402
p=0.7257
p=0.0279
p=0.0502
p=0.7254
p=0.0425
p=0.0151
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

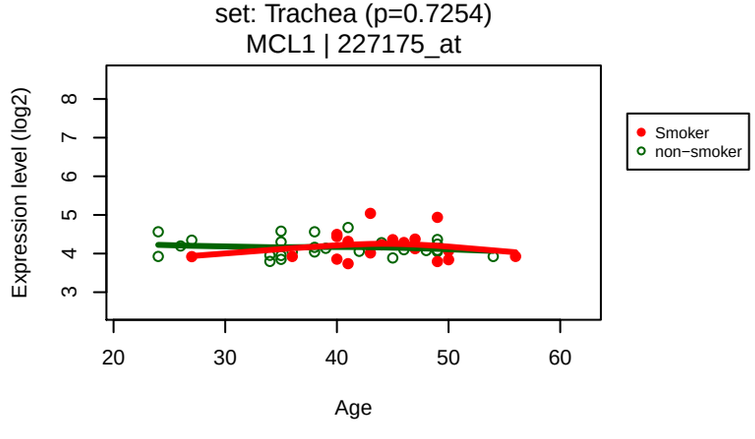

Gene expression ratio in Males
gene: MCL1 | 227175_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2558
p=0.4312
p=0.595
p=0.0502
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MCL1 | 227175_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7005
p=0.1761
p=0.2046
p=0.168
SmallAirway2
LargeAirway2
Trachea
Macrophage2

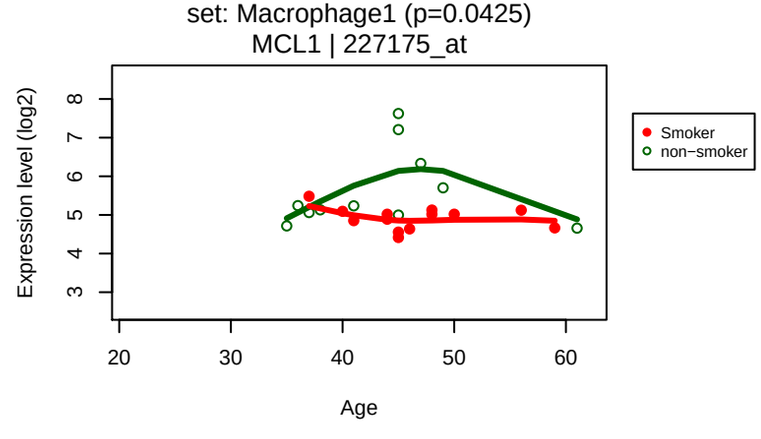

set: SmallAirway1 (p=0.0932)
MCL1 | 227175_at
Expression level (log2)
Age
Smoker
non−smoker

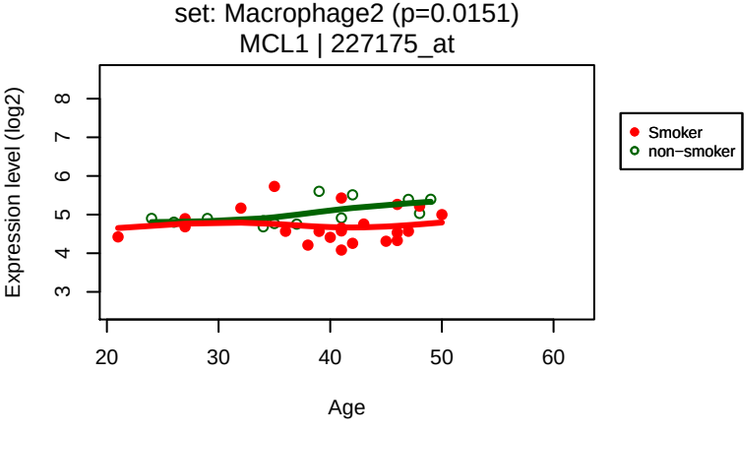

set: SmallAirway2 (p=0.3402)
MCL1 | 227175_at
Expression level (log2)
Age
Smoker
non−smoker

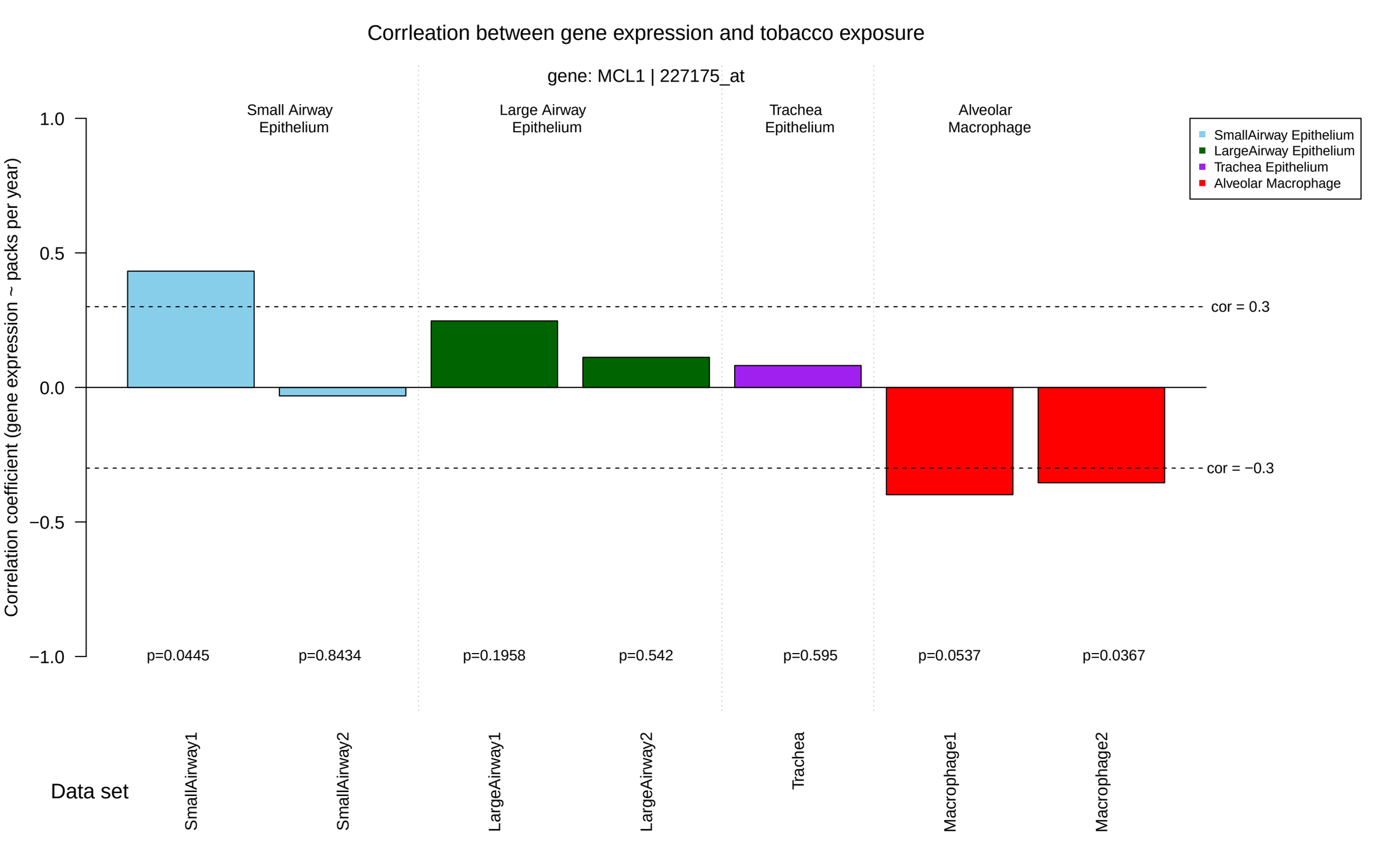

set: LargeAirway1 (p=0.0279)
MCL1 | 227175_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0502)
MCL1 | 227175_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7254)
MCL1 | 227175_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0425)
MCL1 | 227175_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0151)
MCL1 | 227175_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MCL1 | 227175_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0445
p=0.8434
p=0.1958
p=0.542
p=0.595
p=0.0537
p=0.0367
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2